

Data Structures & Algorithms

BCA — Semester III — 2023 — Answer Sheet

Subject Code: CACS201

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group C Attempt any TWO questions[2x10=20]

2. Define stack. Why stack is considered as an ADT? List any four applications of stack.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Evaluate the following postfix expression using the stack: 4 5 + 7 3 - 2 + *

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. What is tower of Hanoi problem? How recursion can be used of Hanoi problem?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Define hashing. Explain how to resolve collisions during hashing using open addressing.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is binary search? Trace the algorithm of binary search to search a key 12 in the data: 11, 19, 5, 2, 7, 21, 8, 21, 12

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is big-oh notation? Explain about divide and conquer strategy with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What are the depth and degree of a node in a tree? Perform pre-order, in-order and post-order traversal of the following tree:

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. How dynamic implementation of the queue can be done? Explain with algorithm. Also explain how insertion and deletion of a node can be done at the end of a singly linked list with algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Define complete binary tree and skewed tree. Write a function to implement heap sort and sort the following data using heap sort: 12, 9, 1, 13, 16, 24, 21, 5

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. How breadth first traversal and depth first traversal can be used for traversing a graph? Explain with example. Use Dijkstra's algorithm to find the shortest path from node A to all other nodes for the following graph.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.